

Lab 5 Reflections

Steven Ford's Reflection

We found missing values, one-hot encoded and feature scaled to deal with NaNs aka missing data. I found this coding a bit more difficult to understand than previous Labs and had to do more searching to work my way through it. An issue I ran into was I couldn't get the age/sex features defined and needed to rerun the panda dataset to get the colab to recognize it. I had been thinking that my code was written wrong but it was the data set somehow had disconnected possibly because of my poor internet connection as I'm doing this with my phone's hotspot on a road trip.

Something new I learned is when I'm trying to write code in colab, it will show a drop down to predict the command and when the code is written out there is a box that talks about what the code is doing. I see that even when trying to write the code without AI, there are still many suggestions and supporters built into this notebook and I assume other notebooks that aid in coding.

I had a little trouble with the last question because I don't remember going over decision tree but I'd assume it's like those charts that you follow by 'yes/then' and I was thinking more about translating that nominal information into numerals but the question was more about feature interaction based on set scales which once I figured that out the feature scaling wouldn't make sense since it's already scaled on its own terms against specified features.

Khin Thu's Reflective Journal

During this lab, I gained a much deeper understanding of the machine learning workflow and the importance of data preprocessing before building a model. Initially, I thought machine learning was mainly about selecting an algorithm and training it on data. However, after completing the notebook, I realized that the quality and preparation of the data are just as important as the model itself. The workflow now appears more logical and structured because each step prepares the data for the next stage.

One of the most important concepts I learned was handling missing values. At first, missing data seemed like a minor issue, but I discovered that many machine learning algorithms cannot process missing values. Working with the Titanic dataset helped me understand why imputation is necessary. I learned how to calculate the median of the Age column and use it to replace missing values. What made this concept click for me was understanding the difference between mean and median. Before this lab, I would have automatically used the average value. However, I learned that the median is more robust when data contains outliers because extreme values do not influence it as much. This showed me how data characteristics can affect preprocessing decisions.

Another valuable lesson was One-Hot Encoding. I initially wondered why categorical values such as "male" and "female" could not simply be left as text. Through the lab exercises, I learned that machine learning models require numerical input. One-Hot Encoding transforms categories into numerical columns while preserving the information contained in the original features. Creating new columns for Sex and Embarked helped me understand how machines interpret data differently from humans. This step demonstrated that data preparation often involves translating human-readable information into a format that algorithms can process efficiently.

Feature scaling was another concept that increased my understanding of machine learning workflows. Before completing this exercise, I did not fully understand why values must sometimes be transformed. By applying StandardScaler to the Age and Fare columns, I observed how large numerical ranges can be standardized. I learned that scaling can prevent features with

larger values from dominating other features in certain algorithms. More importantly, I discovered that not all machine learning algorithms require scaling. Decision Trees, for example, make decisions using threshold-based splits rather than distance calculations. This comparison helped me understand that preprocessing techniques should be selected based on the specific algorithm being used.

One challenge I faced during the lab involved debugging code errors. For example, I encountered a `NameError` because I accidentally typed `"f"` instead of `"df"` when referencing the `DataFrame`. While the mistake was small, it reinforced the importance of carefully reading error messages. Instead of becoming frustrated, I learned to treat errors as part of the learning process. Understanding why an error occurred helped me improve both my coding skills and my confidence in troubleshooting problems independently.

This lab also strengthened the connection between theoretical machine learning concepts and practical implementation. In previous lessons, concepts such as preprocessing, feature engineering, and data cleaning seemed abstract. Through hands-on coding, I was able to see how these concepts are applied in a real workflow. Each step, from handling missing values to encoding categorical data and scaling numerical features, contributed to preparing the dataset for successful model training. This process helped me appreciate that machine learning is not just about algorithms but about creating high-quality input data.

Overall, this lab changed my perspective on machine learning. What initially seemed like a complicated and overwhelming workflow now appears as a sequence of logical steps designed to improve model performance. I learned that careful data preparation is essential for building reliable machine learning systems. The experience also highlighted the importance of critical thinking and problem-solving when working with data. Moving forward, I feel more confident in my ability to prepare datasets, troubleshoot errors, and understand the reasoning behind preprocessing decisions in machine learning projects.

Cristian Delgado's Reflection

Similar to a previous module, all the cells have code on a single line that required restructuring in order to run properly. In order to make things more legible, I used Gemini to organize the code to be more readable. In this module, the coding wasn't as straightforward as maybe the previous ones were, so I utilized Gemini to provide a quick explanation and provide a solution where required. The module itself goes over topics loading and inspecting data, dealing with missing value, encoding, and feature scaling. I had previous experience loading data for some minor ai applications; however, I never bothered to deal with any validation or missing data. I had just assumed everything was ready to go, so this module is giving me insight into what I should do before I utilize any dataset that isn't properly prepared. Additionally, I learned about One Hot Encoding and how it is more or less a requirement for computers to understand and use information. Overall, I gained a better understanding of more of the background processes required to make a dataset useable for the training of a model. I wasn't able to examine the code as much as I had liked due to time constraints working on other assignments, so I will likely review the module once more to make sure I understand some of the code a little better before I continue.

Antonio Garcia's Reflective Journal – Lab 05 Data Preparation

Lab 05 was the first time I really felt the importance of data preparation in a machine learning workflow. Before this module, I understood the idea of training models and making predictions, but I didn't realize how much of the work happens before you even get to the modeling stage. Working with the Titanic dataset made that clear. The data wasn't perfect, and dealing with missing values, categorical variables, and inconsistent formats showed me how essential it is to prepare the dataset properly.

One of the biggest things I learned was why the median is often better for imputing missing values than the mean. The Titanic dataset has a wide range of ages and some extreme fare values. Using the mean could pull the imputed values in a direction that doesn't represent the typical passenger. The median is more stable and less sensitive to outliers, which makes it a better choice for filling in missing ages. Seeing how this small decision affects the dataset helped me understand how preprocessing choices can influence the model later.

Another important part of the workflow was one-hot encoding. Machine learning models can't understand text, so converting categorical variables like Sex and Embarked into numerical columns is necessary. I didn't realize how much this step matters until I saw how the encoded DataFrame changed. Dropping the first category to avoid multicollinearity also showed me that preprocessing isn't just about converting data — it's about making sure the model can interpret it correctly without being confused by redundant information.

Feature scaling was another step that helped me understand workflow complexity. Some models rely on distance calculations or gradient-based optimization, and those models can be affected by differences in scale. Even though decision trees don't need scaling, other models do, so scaling Age and Fare prepares the dataset for multiple algorithms. This made me appreciate how preprocessing keeps your options open and ensures the dataset is ready for whatever model you choose later.

Overall, Lab 05 helped me see how all the pieces fit together in a machine learning workflow. It wasn't just about writing code — it was about understanding why each step matters and how it affects the next part of the process. Even though this is a collaborative project, each of us had to complete our own notebook, and that pushed me to really understand the workflow instead of

relying on someone else's work. Talking with my teammates and comparing our approaches also helped me see different ways to solve the same problems, which made the collaboration feel more like a real data team environment.

This lab showed me that machine learning is not just about algorithms. It's about preparing data carefully, making thoughtful decisions at each step, and understanding how those decisions shape the final model. Lab 05 gave me a clearer picture of the workflow complexity and helped me appreciate how important data preparation is in any machine learning project.